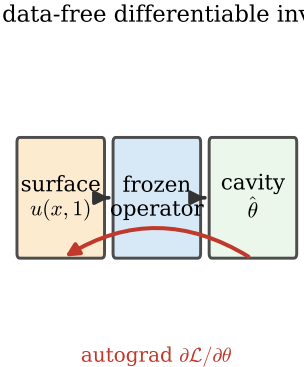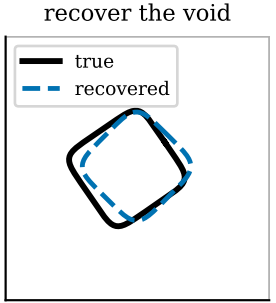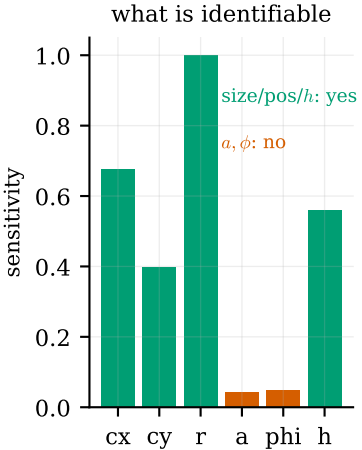Data-free variational operator: forward surrogate + differentiable geometric inverse + identifiability